

Urban Heat Island Effect: Predictive Modelling for Cooling Solutions

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Abstract

As climate change intensifies, cities are becoming increasingly vulnerable to extreme heat amplified by the Urban Heat Island (UHI) effect. This project develops a machine learning framework to predict land surface temperatures across Manhattan, NYC under a range of climate conditions, and to evaluate the cooling potential of green infrastructure interventions. A Random Forest regression model was trained using datasets from NYC Open Data and Landsat 8 thermal and multispectral imagery acquired on August 27, 2024. The model achieved strong performance ($R^2 = 0.747$) in predicting surface temperatures across a 100m grid.

Three future warming scenarios, light ($+1^\circ\text{C}$), medium ($+2^\circ\text{C}$), and harsh ($+3.5^\circ\text{C}$), were simulated both with and without urban degradation (vegetation loss, impervious surface expansion). These projections were compared against scenarios that incorporated green infrastructure, including increased NDVI, expanded tree canopy coverage, permeable pavement, and green roofs applied to the hottest 50% of locations. Across all climate scenarios, green interventions produced substantial cooling benefits of $1.28\text{--}1.90^\circ\text{C}$, with aggressive greening capable of maintaining temperatures near baseline levels under moderate warming.

Feature importance analysis identified impervious surfaces (38%), NDVI (29%), and vegetation metrics (25%) as the strongest predictors of urban heat, aligning with established literature on UHI effect. In conclusion, this study demonstrates a scalable, data-driven approach for predicting urban heat, assessing future risk, and quantifying the cooling potential of greenery to help climate change intervention planning.

1. Introduction

Cities are becoming increasingly vulnerable to extreme heat as climate change accelerates. Urban areas experience the Urban Heat Island (UHI) effect, where dense built environments exhibit significantly higher temperatures than surrounding rural regions. This warming results from heat-absorbing materials such as asphalt and concrete, limited vegetation, reduced airflow between buildings, and heat emissions from vehicles and air conditioning systems.

In many U.S. cities, built environments raise temperatures by more than 8°F during heat waves (The Guardian, 2024). New York City exhibits one of the most pronounced UHI effects among major U.S. cities, with similar patterns observed globally. Cities including Tokyo, Phoenix, Mumbai, and Cairo report severe UHI intensities, with studies documenting temperature differences of up to 5°C in Cairo and as high as 7°C between inland and peripheral zones in Mumbai (Arup, 2023).

The impacts of UHI extend past environmental concerns, reaching social and health implications. Research consistently shows that marginalized neighborhoods often experience the highest temperatures, contributing to health disparities, elevated energy costs, and reduced quality of life (Hsu et al., 2021). Because UHIs reduces nighttime cooling, heat stress becomes more prolonged and dangerous, increasing mortality rates, disrupting sleep, and elevating risks for heat-related illnesses.

This project applies machine learning to model the spatial distribution of urban heat and estimate temperature outcomes under different future climate scenarios. The objectives are to: (1) predict land surface temperatures using urban features such as land cover type, building density, vegetation indices, impervious surfaces, and spatial configuration; (2) simulate three climate warming scenarios with associated changes in land cover; (3) evaluate the cooling benefits of green infrastructure interventions; and (4) identify which features most strongly drive temperature variation.

Overall, this work represents both a prediction and inference problem: forecasting continuous land surface temperature across Manhattan while identifying the key factors that drive urban heat up. Past modeling, the analysis estimates how interventions such as tree planting, green roofs, and permeable materials could alter temperature outcomes in specific locations. The results aim to inform others about climate change and urban adaptation planning.

2. Data Sources

This analysis integrates three publicly available geospatial datasets for Manhattan, NYC, spatially aligned to a 100m × 100m grid covering 19,800 points.

NYC Building Footprints

Source: NYC Open Data Portal

Features: Building footprints, estimated heights, density per 100m grid

Temporal: Current building stock (2024)

NYC Street Tree Census

Source: NYC Open Data Portal

Features: Tree locations, species, diameter, estimated canopy coverage

Temporal: 2015 tree census

Landsat 8 Satellite Imagery

Source: USGS Earth Explorer (Landsat Collection 2)

Scene ID: LC08_L2SP_013032_20240827 (Path 013, Row 032)

Acquisition Date: August 27, 2024

Bands Used: Band 10 (Thermal), Band 4 (Red), Band 5 (NIR)

Features: Land Surface Temperature (LST), NDVI, NDBI, impervious surface estimation

Resolution: 100m thermal bands, resampled to 100m analysis grid

3. Methodology

Feature Construction

Urban, vegetation, and surface characteristics were derived from NYC building, tree datasets and Landsat 8 imagery. These features were computed for each 100 m grid cell.

Building Features:

1. Building density (buildings per km²)
2. Average building height (feet)
3. Building volume density
4. Urban density index (density $\times$ height / 1000)

These features show structures which influence heat retention and airflow.

Vegetation Features:

1. NDVI (Normalized Difference Vegetation Index, -1 to +1)
2. Tree canopy coverage in percentage
3. Tree density (trees per km²)
4. Vegetation score

These features represent cooling capacity and evapotranspiration, which is the process of water moving from the Earth's surface to the atmosphere, combining evaporation from soil and water surfaces with transpiration from plants.

Surface Features:

1. Impervious surface fraction (0-1)
2. NDBI (Normalized Difference Built-up Index)

These features represent heat absorbing materials and built-up intensity.

Predictive Model

A supervised machine learning approach was used to predict land surface temperature for each grid cell.

- **Algorithm:** Random Forest Regressor
- **Model parameters:** 100 trees, maximum depth of 15
- **Train/test split:** 80/20
- **Random state:** 42 for reproducibility
- **Target variable:** Land Surface Temperature (LST) from Landsat 8 Band 10 (August 27, 2024), calibrated using Landsat Collection 2 algorithms.

Model Performance

The model achieved:

- **R²:** 0.747
- **RMSE:** 2.55°C

This level of performance is considered strong for urban heat modeling, as urban temperature varies complexly over short spatial scales.

Climate Scenarios

Three future warming scenarios were designed to reflect increasing climate stress and land degradation patterns:

Scenario	Warming	Vegetation Loss	Impervious Increase
Light	+1.0°C	3%	2%
Medium	+2.0°C	5%	4%
Harsh	+3.5°C	8%	6%

For each scenario, model inputs were adjusted by:

- Increasing the LST baseline according to scenario warming
- Reducing vegetation features (NDVI, vegetation score, tree canopy)
- Increasing impervious surface

These changes simulate conditions expected under climate stress and urban expansion.

Green Infrastructure Intervention

Each climate scenario was simulated with and without green infrastructure. Green infrastructure was applied to the hottest 50% of grid cells (9,895 locations), identified using baseline temperature predictions.

Intervention Methods

1. In hot areas
 - NDVI increase: +0.30
 - Added tree canopy coverage: +25%
 - Impervious surfaces reduced by 20%
2. Green roofs on buildings >100 ft (7,753 locations)

- NDVI increase: +0.15
- Added canopy equivalent: +12%

These interventions represent strategies including tree planting, permeable pavements, pocket parks, and widespread green roof adoption.

Scenario Evaluation

For each grid cell and scenario, the adjusted feature set was passed through the trained Random Forest model to generate LST projections. Results were summarized citywide and used to quantify:

- Temperature changes under climate warming
- Cooling benefits are attributable to green interventions
- Spatial patterns of temperature amplification or mitigation

This framework allows exploration of how land cover, vegetation change, and climate stress combine to shape future urban heat conditions.

4. Results

Baseline Conditions

Current Mean Temperature: 36.32°C (August 2024, Landsat LST)

Standard Deviation: 5.1°C

Temperature Range: 19.3°C to 54.7°C

Spatial Pattern: Higher temperatures in dense commercial districts (Midtown, Financial District), cooler in parks (Central Park, Riverside Park)

Note: LST (Land Surface Temperature) are higher than air temperature, especially on sunny days. August 27, 2024 was a clear, warm day in NYC.

Climate Scenario Projections

Scenario	No Greenery	With Greenery	Cooling Benefit
Baseline	36.32°C	-	-
Light	37.30°C (+0.99°C)	35.41°C (-0.91°C)	-1.90°C
Medium	38.07°C (+1.76°C)	36.41°C (+0.09°C)	-1.67°C
Harsh	39.19°C (+2.87°C)	37.91°C (+1.59°C)	-1.28°C

Key Findings

1. Warming without intervention results in temperature increases of 0.99–2.87°C, depending on scenario.
2. Green infrastructure produces substantial cooling of 1.28–1.90°C across all scenarios.

3. Greatest benefits occur under lighter warming, when vegetation remains more resilient and interventions can more effectively offset heat stress.
4. Aggressive greening stabilizes temperatures, keeping mean LST near baseline even under medium warming.
5. Citywide benefits emerge from targeted actions: cooling the hottest 50% of the grid reduces heat island intensity throughout Manhattan.

Feature Importance

Top drivers of urban heat identified by the Random Forest model:

Impervious surfaces (38.0% importance) - highest predictor
NDVI (29.1% importance) - vegetation index
Vegetation score (25.2% importance) - composite vegetation metric
Urban density index (2.7% importance)
Average building height (1.8% importance)

Interpretation: Surfaces that absorb and retain heat (pavement, buildings) are the strongest predictors, while vegetation provides the strongest cooling effect. Together, vegetation-related features account for 54.3% of model importance.

5. Challenges

Interpretation

The results demonstrate that:

Green infrastructure is an effective climate adaptation strategy even under significant warming.
Vegetation and permeable surfaces have measurable cooling effects at the city scale.
Targeted interventions in the hottest areas provide city-wide benefits.
Early intervention (light scenario) provides greatest per-degree cooling benefit.

Limitations

This project came with several challenges that affect how the results should be interpreted.

Data challenges:

- The street tree data is from 2015, while the satellite data is from 2024, so the number and size of trees may have changed.
- Building heights had to be estimated because detailed height data like LiDAR was not available.

- The analysis is based on a single satellite image from one summer day, which does not show how temperatures change across seasons or years.
- The 100-meter grid size is large and can miss trivial details like shaded streets or small patches of greenery.
- Land surface temperature is different from air temperature, so results cannot be compared directly to weather station readings.

Model challenges:

- There were no ground sensors available to check how accurate the model's temperature predictions were.
- The green infrastructure scenarios had to be simplified. In real life, trees, plants, and green roofs all behave differently depending on their type and condition.
- The model gives static snapshots, it doesn't show how heat might change over time or respond to feedback loops in the city.

Scenario challenges:

- The warming and land-cover changes were applied evenly across Manhattan, even though real-world changes would be uneven.
- The scenarios do not include things like storms, humidity changes, or extreme weather.
- The model does not consider cost, space, politics, or practical issues that might limit the amount of greenery that can be added.

Even with these challenges, the results still give a useful picture of how surface materials and vegetation affect heat across the city.

Conclusion

This project used machine learning and several NYC datasets to predict land surface temperatures in Manhattan and test how heat might change under different future climate conditions. The model performed well and showed that surfaces like concrete and asphalt make the city hotter, while vegetation helps it cool down.

Across all warming scenarios, adding green infrastructure, such as trees, green roofs, and permeable pavement, reduced temperatures by 1.28–1.90°C. In lighter warming scenarios, these interventions were strong enough to keep temperatures close to today's levels. These results show that green infrastructure can make a meaningful difference, especially if cities act early.

Overall, this project provides a simple framework for understanding where heat risks are highest and how greenery can help reduce them. The approach could be applied to other cities to support planning and guide future climate adaptation efforts.

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